DARSHAN S

AI/ML Engineer
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GitHub Portfolio LinkedIn

Professional Summary

AI & ML undergraduate with experience building computer vision systems for geospatial property analysis and document-grounded RAG applications. Skilled in Python, FastAPI, React, semantic caching, intelligent LLM routing, model evaluation, and full-stack development. Focused on developing practical AI systems with measurable cost optimization, reliable retrieval, and observability.

Education

M S Ramaiah University of Applied Sciences — In Progress
B.Tech, Artificial Intelligence & Machine Learning — CGPA: 8.0

Christ Junior College — 2023
Pre-University Course, KSEAB, Karnataka — 83.6%

Cambridge Public School — 2021
10th Boards, CISCE — 92.8%

Experience

MPRO9 — Jul. 2026 – Oct. 2026
AI Intern – Project Development & Implementation — *Mysore, India*

- Built GeoVision AI for property identification and analysis using geospatial and visual data.
- Developed YOLO-based object detection workflows for analyzing 360-degree imagery.
- Integrated visual data with GPS information using Insta360 X4 imagery and u-blox GPS modules.
- Contributed to data preprocessing, model development, testing, and optimization.

Personal Projects

DECIPHER – Meeting-to-Execution Agent
Next.js, React, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Zod, Google Gemini API, OpenRouter, Trello API, Vercel

- Built an AI-powered meeting intelligence platform that turns meeting transcripts into structured decisions, commitments, risks and dependencies, each linked back to the transcript line it came from.
- Designed human-in-the-loop approval workflows: AI output is validated against the transcript and reviewed by a person before anything is saved or sent to Trello.
- Added cross-meeting decision intelligence with a Decision Timeline, Commitment Tracker, Impact Analysis and AI Command Center, plus Trello integration for approved commitments.

Cost-Aware RAG – Semantic Caching & Intelligent Model Routing
Python, FastAPI, React, TypeScript, SQLite, Groq, Sentence Transformers

- Built a document-grounded RAG pipeline with PDF/TXT ingestion, page-aware chunking, vector retrieval, source citations, and insufficient-evidence handling.
- Implemented evidence-aware semantic caching and complexity-based LLM routing, reducing estimated inference cost by 59.2% across 38 requests in offline evaluation.
- Created isolated temporary visitor workspaces using server-issued tokens and in-memory SQLite, with session-scoped documents, retrieval, caching, and telemetry.

Smart Traffic Vehicle Counter
Python, YOLO11n, ByteTrack, OpenCV, Streamlit

- Developed a vehicle detection, tracking, classification, and counting system supporting four vehicle classes, with direction and speed estimation.
- Fine-tuned YOLO11n on 500 UVH-26 images, achieving 77.7% precision, 69.3% recall, 77.1% mAP@50, and 60.3% mAP@50–95 on a 296-image validation sample.
- Integrated ByteTrack, trajectory analysis, and line-crossing logic into a Streamlit analytics dashboard; benchmarked ByteTrack against BoT-SORT (43 vs. 75 track IDs).

AI Health Tracker
React.js, Node.js, Express.js, MongoDB, Google Gemini API, JWT

- Built a full-stack healthcare platform for health-record management and personalized AI-generated insights.
- Implemented JWT authentication, REST APIs, MongoDB persistence, and a responsive React dashboard.

Technical Skills

Languages: Python, JavaScript, Java, C
Frontend: React.js, TypeScript, HTML5, CSS3, Tailwind CSS
Backend: FastAPI, Node.js, Express.js, REST APIs
AI/ML: RAG, Semantic Caching, LLM Routing, Sentence Transformers, TensorFlow, Scikit-learn, YOLO, Computer Vision, NLP, Deep Learning, LangChain, LangGraph, Google Gemini API
Databases: SQLite, MongoDB, SQL
Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook, Pytest, OpenCV, Streamlit
Hardware: Insta360 X4, u-blox GPS
Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Operating Systems, Computer Networks

Certifications

- Oracle Certified: AI Vector Search Certified Professional
- AI Fluency Framework and Foundations – Anthropic
- AI Literacy – IBM
- AI Fundamentals with IBM SkillBuild – Cisco
- Data Analytics Essentials – Cisco
- Prompt Design in Vertex AI – Google
- Generative AI Foundations – upGrad in collaboration with Microsoft